

7. Edinburgh Space-Time Geostatistics Users Guide

Running Title: Spatio-Temporal Geostatistical Methods

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7.1 Declaration

I am the sole author of all text and program code described in the following chapter, although the text draws heavily from sources referenced in the text. All references are cited, and I declare the text to be my own work. Fortran 90 code for the programs described is included in electronic format in an appendix.

7.2 Abstract

We document a set of Fortran 90 programs (the *Edinburgh Space-Time Geostatistics Package*) for the spatio-temporal regionalization of data via geostatistical methods. The programs utilize the product and product sum covariance representations of spatio-temporal data interactions. The code described allows interpolation of a data set over and arbitrarily spaced grid in continuous spatial and temporal coordinate systems. Error analyses are provided via the Jack Knife. The resultant spatio-temporal fields represent the expectation of a random function (RF), conditioned on the observed data and covariance model. The techniques implemented allow production of fields of estimation variance. We also provide code for simulation from the RF, via Sequential Gaussian Simulation (SGS). The SGS technique makes random draws from the RF, and allows the user to quantify the uncertainty of the interpolated field. SGS allows *Monte Carlo* analysis of the regionalisation by ensuring draws from the distribution (described by the expectation and estimation variance) conform to the observed spatio-temporal covariance of the data. The SGS technique is particularly useful in cases where the user intends to parameterise a model with a regionalised field, as the interpolation uncertainty can be propagated through the model to produce appropriate confidence intervals. Instructions for the use of the software are provided, along with sufficient background theory to successfully implement spatio-temporal regionalisation. A section discussing practical aspects of geostatistical modelling is also provided as an aid to first time users.

7.3 Introduction

The Edinburgh Space-Time Geostatistics software is a collection of three programs for the purpose of upscaling observations of a variable of interest to a set of coordinates distributed over some region of interest. The programs allow upscaling of data using the inverse distance weighting method (*IDW.exe*) and Kriging (*Geostats.exe*). Use of the Kriging algorithm generates estimates at unsampled locations using a random function model, specified from the data distribution and a description of the spatio-temporal relationship between observations. Simulations can be drawn from this model using the program *Gaussim.exe*.

The field of geostatistics has been in development since the late 1960s (Cressie 1990), and in the last decade, geostatistical techniques have expanded to encompass spatio-temporal estimation problems (Kyriakidis and Journel 1999, Christakos 2000, De Cesare et al. 2001b, De Iaco et al. 2001). While multi-dimensional implementations existed prior to this, unique challenges of spatio-temporal implementations remained: In particular, describing the space-time autocorrelation structure of the data was problematic (Kyriakidis and Journel 1999, De Cesare et al. 2001a, Gneiting 2002b, Gneiting et al. 2005). In 2001, De Iaco et al. introduced the product-sum covariance model (De Iaco et al. 2001), allowing intuitive and simple construction of spatio-temporal descriptions of autocorrelation, and provided Fortran 77 code for estimation via these techniques (De Cesare et al. 2002).

A wealth of software exists for geostatistical estimation (e.g. Deutsch and Journel 1998, De Cesare et al. 2002) prompting the question, why develop a new set of tools at all? The intention of the current software and documentation is to provide a user friendly and flexible implementation of spatio-temporal geostatistical methods as

described by De Cesare et al. (2001a) with custom built Fortran 95 software, rather than the modified GSLIB (Deutsch and Journel 1998) code implemented by De Cesare et al. (2002): In particular, we wished to allow cross-validation of entire time series, rather than the ‘one observation at a time’ method implemented by De Cesare et al. (2002), and to provide additional outputs not provided by the GSLIB code.

Theoretical introductions to geostatistical techniques are abundant (Clark 1979, Isaaks and Srivastava 1989, Cressie 1991, Goovaerts 1997, Deutsch and Journel 1998, Wackernagel 1998), and relevant material for spatio-temporal implementations can be found in (Kyriakidis and Journel 1999, De Cesare et al. 2001b, a, De Iaco et al. 2001, Heuvelink and Webster 2001, De Cesare et al. 2002, Gneiting 2002b, De Iaco et al. 2003, Gneiting et al. 2005). It is not our intention to re-cover this material: Whilst we provide a brief introduction to the relevant material necessary to successfully implement the techniques described, we assume some familiarity with the theory behind Kriging, and focus on a more applied description of spatio-temporal estimation techniques as implemented in the accompanying software, in the hope that they may prove useful to others. The program is supplied with the GNU general public license agreement: Please acknowledge the authors when using this software.

7.4 The Random Function Model and the Requirement of Stationarity

Linear geostatistical methods employ a probabilistic approach to upscaling; each data point is conceived of as a draw from a normal distribution, referred to as Gaussian random variable (RV). This collection of RVs are related to each other by some

quantifiable distance metric. This collection of spatio-temporally dependent random variables is known as a random function.

Kriging produces an estimate by calculating the conditional expectation of the random function (RF) for the estimation location, given the observed values. The expectation of the RF is fully defined by the expectation and covariance structure of the RF. Local spatial uncertainty can be explored by repeated draws from the RF *via* sequential simulation (see section 7.9.1.iv). Generally, the statistical description of autocorrelation is provided by the semivariogram, although other choices are possible (see section 7.5.2). We generally infer the covariance structure from the semivariogram.

In order to make inferences about any distributional characteristic (expectation, variance *etc.*) of an RV it is useful to have repeated measurements from its distribution. Therefore, in order to estimate the autocorrelation between (the RVs representing) observations separated by distance h , it is useful to have multiple observations on their joint distribution: Such a set of observations are never truly available, since subsequent samples at any location will be drawn at different temporal coordinates.

In the absence of repeated measures, we may substitute spatial replication for *in situ* repetition. The intention is that by grouping together all observation pairs separated by distance h ($\pm$ some lag tolerance), we can reconstruct the RVs, providing the following condition is met: To ensure that the description of autocorrelation is readily calculable, we require that all RVs in lag h share same distribution. This implies that the mean and variance of the data must be translation invariant (homogenous) across the region of interest, a condition referred to as second order stationarity. However it is sufficient that the mean and variance are homogenous only within the lag h ; referred to as intrinsic stationarity. If this condition is satisfied, the similarity of data pairs can be defined purely in terms of their separation distances, greatly simplifying the specification

of the RF. If the underlying function is not intrinsically stationary, it may be possible to model a regionalisation using the relative semivariogram (Cressie 1985b). Further discussion can be found in (Cressie 1991, Goovaerts 1997, Deutsch and Journel 1998).

In order to satisfy second order stationarity in practice, it is often necessary to remove trends from the observations via some model. These trends can either be removed from the data and added back to interpolated estimates of the residuals, or incorporated directly in the interpolation scheme (see sections 7.8.2.i and 7.8.2.iii).

7.5 Initial Data Modelling

7.5.1 Accounting for Global Trends

Geostatistical methods model a RF as a combination of the mean m and a spatially coloured, normally distributed noise process ε . The mean is not necessarily uniform across the domain, and may be defined by some global trend function, whilst ε is defined by a model of autocorrelation. This decomposition of the process into a large-scale trend component and small-scale autocorrelated noise is familiar from time series analysis (Cressie 1991), and forms the rationale of non-stationary geostatistics.

Large-scale trends in the data may cause problems in estimating a model for ε , and may often cause statistics such as the semivariogram to become unstable, tending to infinity as separation distances become large (see section 7.5.2.i). On the other hand, large-scale trends may provide another source of information for constraining estimates, and may provide realistic physical dependencies in the resulting estimates. Furthermore, it is conceptually useful to partition data variation between known, measured trends and dependencies, and a stochastic but structured error component; this latter component reflects the unknown or unknowable latent variables which are either unmeasured, or

effectively too complex to control for in data collection. In either case, it is necessary to account for these trends and appropriately quantify them. Large-scale trends can be effectively modelled either through process-based knowledge of the variable of interest, or empirically through the use of trend surface analysis (Haining 2003) or the median polish (Tukey 1977), as popularised by Cressie (Cressie 1984, Cressie 1991).

All three programs provided in the Edinburgh Space-Time Geostatistics package allow for incorporation of large-scale trends in the estimation procedure. When large-scale trends are present in the data, it is necessary to remove them prior to the calculation of the semivariogram. If the trends are related to secondary variables that are known across the study site, non-stationary methods of geostatistics may be applicable (see sections 7.8.2.i and 7.8.2.iii). However, in the case of linear longitudinal trends or similar, non-stationarity may be accounted for by appropriate choice of kriging neighbourhood (see section 7.8.2.ii), providing local stationarity is achieved for the conditioning data (see Journel and Rossi 1989 for further details).

7.5.2 Accounting for Autocorrelation

7.5.2.i *The semivariogram*

The semivariogram is a method for summarizing the pattern of spatial or temporal variation (autocorrelation) of an observed phenomenon (Hudson and Wackernagel 1994, Gringarten and Deutsch 2001); describing the way in which similar observation values are clustered in space or time, in accordance with Tobler's first law of geography (Tobler 1970). The semivariogram is therefore a measure of the *dissimilarity* of data pairs as the separation between them increases, and is essentially the inverse of the auto-covariance of the data (Deutsch and Journel 1998).

The term semivariogram is often loosely applied to describe a whole series of possible statistics applied to a set of observations with attached coordinates, but most correctly it applies to Matheron's 'classical estimator' (Matheron 1962) of autocorrelation: For a set of n observations, we may choose any pair of data values, z_i and z_j with associated coordinates u_i and u_j . The dissimilarity of the data pair can then be calculated as half the squared difference between the observed values:

$$\gamma_{ij} = \frac{[z(u_i) - z(u_j)]^2}{2} \quad (7.1)$$

Plotting the differences γ_{ij} against separation distance results in the variogram cloud. The variogram cloud is typically diffuse, and suffers from pointwise instability (Diggle et al. 2002). By averaging the pairwise dissimilarities over a set of n spatial lags $H = \{h_1, \dots, h_n\}$ we arrive at the more stable semivariogram, denoted $\hat{\gamma}(h)$:

$$\hat{\gamma}_h = \frac{1}{2N(h)} \sum_{i=1}^{N(h)} [z(u_i) - z(u_i + h)]^2 \quad (7.2)$$

This is actually the first moment of inertia of the lag (Isaaks and Srivastava 1989), reflecting the width of the scatter of z_u on z_{u+h} from the 45° line, where u is any location in the region of interest and h is a separation vector.¹

Goovaerts (1997) points out that a generalised semivariogram estimator is possible by changing the power of equation 7.2 from 2 to ω :

$$\hat{\gamma}_h = \frac{1}{2N(h)} \sum_{i=1}^{N(h)} [z(u_i) - z(u_i + h)]^\omega \quad (7.3)$$

¹ **NOTE:** The semivariogram can be used to derive an estimate of the fractal dimension of a process, which relates to the surface roughness of the RF. Fractal dimension can be calculated from the slope of the linear portion of the log-log plot of semivariogram (Palmer 1988, Leduc et al. 1994). An interesting extension is provided in (Gneiting and Schlather 2004).

For $\omega = 2$ we arrive at the ‘classical’ estimator, but by reducing the value of ω we can reduce the influence of extreme values on the measure $\hat{\gamma}_h$. Values of note are $\omega = 1$ and $\omega = 0.5$, returning the mean absolute deviations, and square root deviations (known as the *madogram* and *rodogram* respectively), and are commonly used alternatives (Deutsch and Journel 1998). These more robust alternatives may help to make inferences about the nature of the autocorrelation of a process when used in conjunction with the semivariogram.

Providing the mean and variance of the observed phenomenon is translation invariant (i.e. identical for any subregion of the domain), the semivariogram is a valid description of the spatial autocorrelation structure of the data (Cressie 1991). If the above condition is satisfied, the difference between any pair of data points at arbitrary separation is purely a function of their separation vector h , and the phenomenon is referred to as a stationary process.

For a stationary process, we generally observe an increase in semivariance with increased separation vector h , up to some threshold distance, referred to as the range. At separation distances greater than the range, the semivariance remains at a constant ‘sill’ value.

In the case of large-scale trend structures (for example, an East-West gradient), a parabolic structure is commonly observed (Clark 1979), where the semivariance values rise in an unbounded fashion. In this case, the phenomenon is said to be non-stationary (Deutsch and Journel 1998), and some method of trend removal is necessary (see section 7.5.1), before the residual values can be analyzed.

In many cases, the semivariogram will display a discontinuity at the origin. This behaviour is commonly referred to as a ‘nugget effect’, as it reflects the condition where values of the observation vary abruptly at the microscale. This term was coined in

mining geostatistics, where nuggets contained in samples from gold grades tended to produce this effect. Discontinuous behaviour of the semivariogram at the origin can usually be accounted for by a combination of microscale variability below the resolution of the sampling equipment, and sensor noise.

7.5.2.ii The Covariogram

The covariogram is the covariance between data pairs at each lag. By convention, the first value in the comparison is referred to as the 'tail' value, and the second value in the pair is called the 'head' value [*tail* $\xrightarrow{h}$ *head*] (Goovaerts 1997, Deutsch and Journel 1998):

$$c(h) = \frac{1}{N(h)} \sum_{i=1}^{N(h)} z(u_i) \cdot z(u_i + h) - m(-h) \cdot m(+h)$$

where

$$m(-h) = \frac{1}{N(h)} \sum_{i=1}^{N(h)} z(u_i) \tag{7.4}$$

$$m(+h) = \frac{1}{N(h)} \sum_{i=1}^{N(h)} z(u_i + h)$$

Where $m(-h)$ and $m(+h)$ are the tail and head value means respectively (lag means).

Whilst the semivariogram is the most common description of autocorrelation used in the practice of geostatistics, in general the covariogram is used in the kriging algorithm, because of difficulties in using the semivariogram representation for simple kriging, and improvements in computational efficiency (Goovaerts 1997). The covariogram is readily obtained from the semivariogram by subtracting the semivariogram from the sill variance (see section 7.5.2.i).

Unbounded models have no covariance counterpart; in this case pseudo-covariance is calculated by subtracting the semivariogram from a sufficiently large constraint, such that the resulting value ≥ 0 .

7.5.2.iii The Correlogram

In some cases we may wish to discuss data dependency in terms of autocorrelation, as is common in time series analysis. The correlogram is defined by the correlation between the data values in each lag. It is easily obtained from the covariogram by standardisation:

$$\rho(h) = \frac{c(h)}{\sqrt{\sigma^2(-h)\sigma^2(+h)}}$$

where

$$\sigma^2(-h) = \frac{1}{N(h)} \sum_{i=1}^{N(h)} [z(u_i) - m(-h)]^2 \quad (7.5)$$

$$\sigma^2(+h) = \frac{1}{N(h)} \sum_{i=1}^{N(h)} [z(u_i + h) - m(+h)]^2$$

Where $\sigma^2(-h)$ and $\sigma^2(+h)$ are the variances of the tail and head values respectively (lag variance).

7.6 Continuous Models of Autocorrelation

Having established the pattern of autocorrelation by calculation of the semivariogram, it is desirable to express this structure in continuous terms. Modelling of the semivariogram is necessary for geostatistical estimation, and allows inference to be drawn on various properties of the autocorrelation, such as its effective range and asymptotic variance (known as the sill).

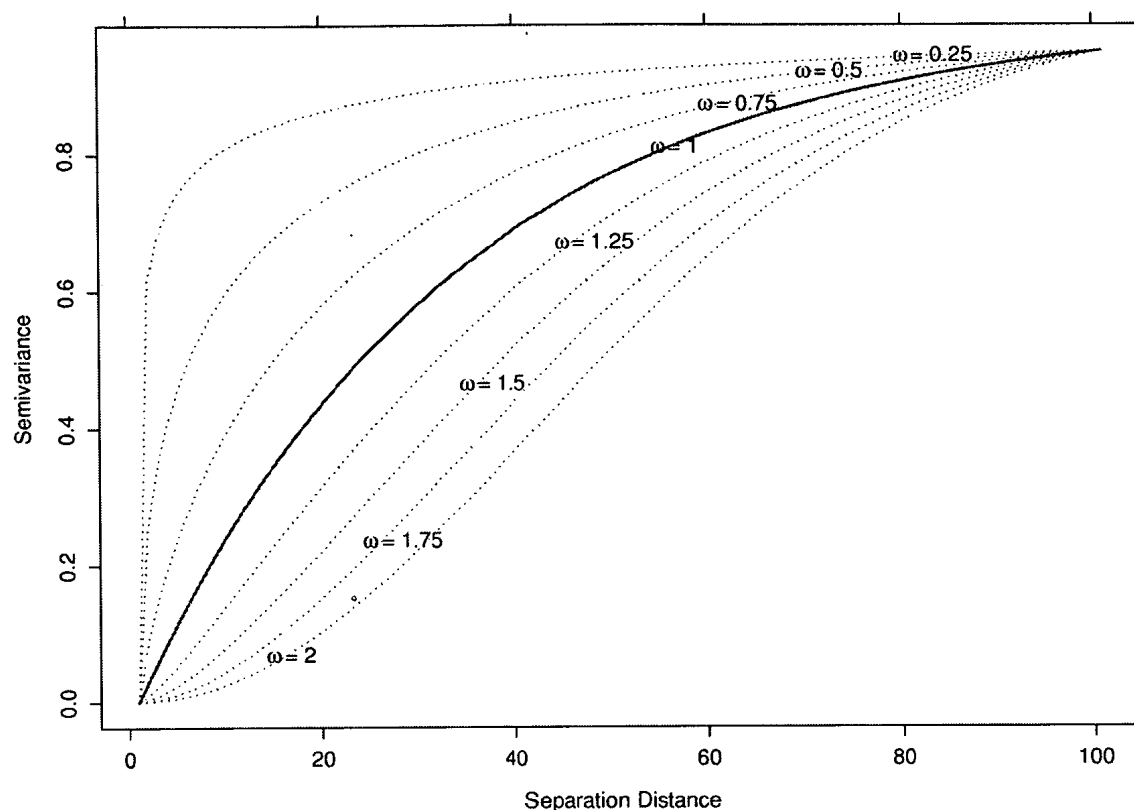


Figure 7.1 Powered exponential model with varying smoothness parameter ω . As ω increases semivariance increases more slowly at the origin, indicating a very continuous process.

The semivariogram can be modelled with any conditionally negative definite function (Mcbratney and Webster 1986). Rather than testing proposed models for permissibility (e.g. Christakos 1984), it is usual to use one of a set of basic models that are known to be permissible. There are many such models in common use (see Deutsch and Journel 1998), and these basic model structures may be combined in a linear manner to form complex models (Goovaerts 1997).

7.6.1 Permissible Semivariogram Models

The following models are available in the Edinburgh Space-Time Geostatistics programs: Powered exponential, Gaussian, Spherical, Rational quadratic, Power, Hole

effect and dampened hole. These models provide a wide range of functional forms reflecting different surface characteristics. Details of their use and limitations are presented below:

- ***Powered Exponential Model***

$$\hat{\gamma}_h = 1 - \exp\left[-3\left(\frac{h}{\phi}\right)^\omega\right]$$

where

$$0 < \omega \leq 2 \quad (7.6)$$

Where ω is a smoothness parameter (Figure 7.1). Note that $\omega=0$ is not a valid model. When $\omega=2$ the model is equivalent to the Gaussian model presented below.

- ***Gaussian Model***

$$\hat{\gamma}_h = 1 - \exp\left(\frac{-3h^2}{\phi^2}\right) \quad (7.7)$$

The Gaussian model reflects a very smoothly varying process. It may exhibit unstable behaviour, and is not recommended for use without a nugget effect.

- ***Spherical Model***

$$\hat{\gamma}_h = \begin{cases} 1.5\frac{h}{\phi} - 0.5\left(\frac{h}{\phi}\right)^3 & \text{if } h \leq \phi \\ 1.0 & \text{if } h > \phi \end{cases} \quad (7.8)$$

The spherical model is widely used in the literature. It provides a model with almost linear behaviour near the origin, which abruptly levels to the sill value.

- ***Rational Quadratic Model***

$$\hat{\gamma}_h = h^2 \div \left(\frac{1 + h^2}{\phi} \right) \quad (7.9)$$

The rational quadratic model is roughly sigmoidal with moderate smoothness near the origin, having a shorter left hand tail than the powered exponential. It displays linearity at low to intermediate ranges, with a pronounced smooth shoulder towards the sill. Despite its somewhat similar appearance to powered exponential models with high values of ω , the rational quadratic model displays less smoothness at short range and greater stability than the powered exponential.

- ***Power Model***

$$\hat{\gamma}_h = h^\omega \quad (7.10)$$

Where ω is the power law, describing the rate of decay in autocorrelation as distance increases; lower values therefore imply a smoother interpolated surface. The power model has no covariance counterpart, as it is unbounded ($\hat{\gamma}_h \rightarrow \infty$ for large h): We implement the model using a pseudo-covariance counterpart (see section 7.5.2.ii). Given that parabolic semivariograms indicate non-stationarity (see section 7.5.2.i), we do not recommend the use of the power model, and include it only for completeness; we recommend de-trending the data (see section 7.5.1) prior to semivariogram modelling to avoid its use.

- ***Hole Effect (Cosine) Model***

$$\hat{\gamma}_h = 1 - \cos\left(\frac{h}{\phi}\pi\right) \quad (7.11)$$

The hole effect describes a periodicity in the underlying features, such as seasonality or diurnal cycling. The range ϕ describes the wavelength of the periodicity. The hole effect is intended for use in nested models (see section 7.6.1.ii). In order for the resulting semivariogram to be valid (positive definite), the hole effect can only be applied in one direction: Therefore its use in time is straightforward. To implement the hole effect spatially, the user should specify an appropriate geometric anisotropy ellipse (see section 7.6.1.iii), with a very large range in the direction perpendicular to the periodicity.

- ***Dampened Hole Model***

$$\hat{\gamma}_h = 1 - \exp\left(\frac{-3h}{d}\right) \cdot \cos\left(\frac{h}{\phi}\pi\right) \quad (7.12)$$

The dampened hole effect model decreases the level of periodicity as distance increases. Again, the range ϕ describes the wavelength of the periodicity, whilst parameter d defines the distance at which 95% of the periodicity is removed from the signal.

In all cases h is the lag distance and ϕ is the range over which the data exhibit autocorrelation. Functional forms for all models are presented in Figure 7.2. For the

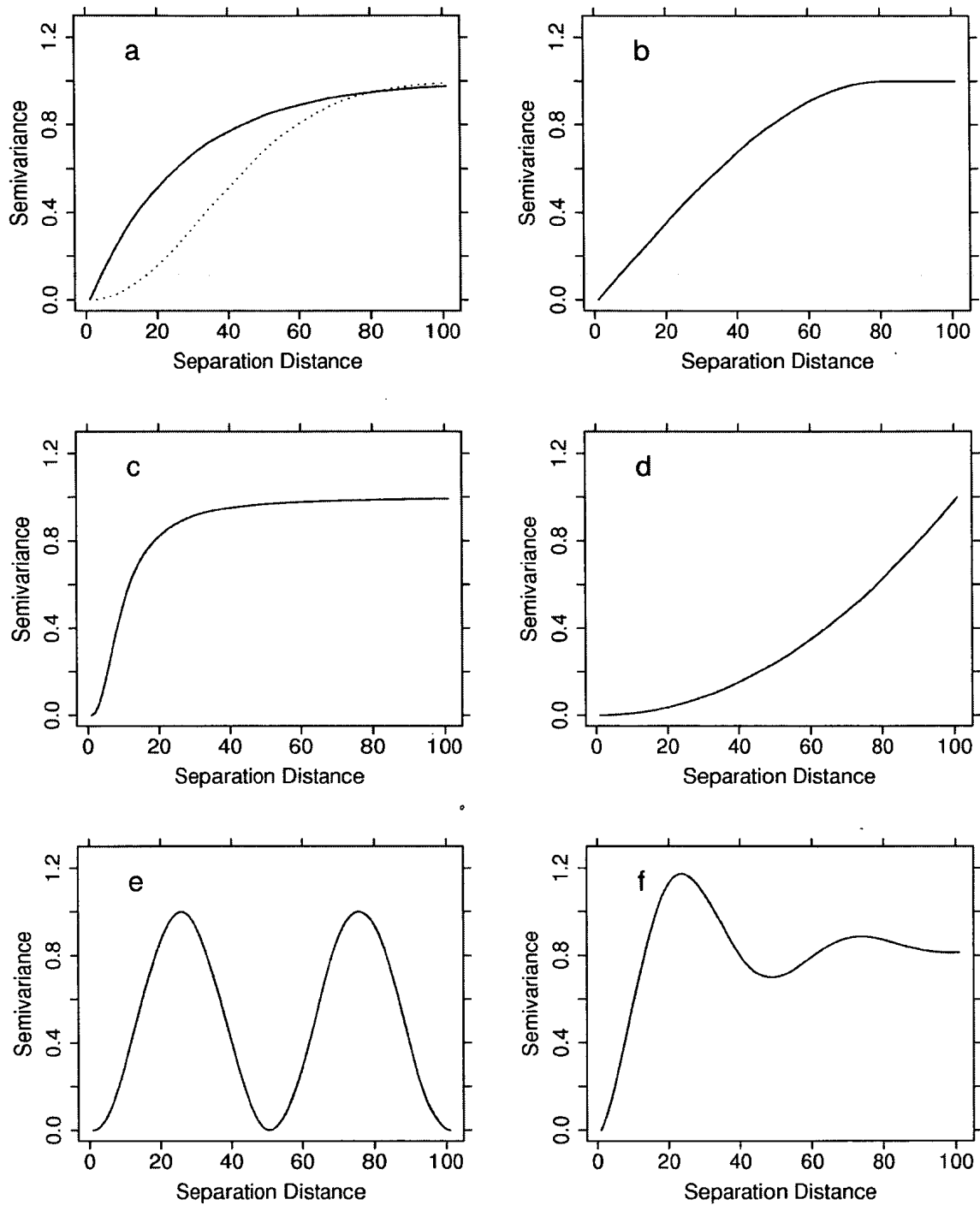


Figure 7.2 Semivariogram models available in Edinburgh Space-Time Geostatistics. (a) Powered exponential model (equation 7.6). The solid line illustrates $\omega=1$, whilst the broken line illustrates $\omega=2$, equivalent to a Gaussian model (equation 7.7). (b) Spherical model (equation 7.8). (c) Rational quadratic (equation 7.9). (d) Power model (equation 7.10). (e) Hole effect (equation 7.11). (f) Dampened hole (equation 7.12).

exponential and Gaussian models ϕ is the effective range, interpreted as the distance at which semivariance reaches 95% of the asymptotic ‘sill’ variance: The factor of 3 in the numerator solves for effective range². The models are rescaled with a contribution parameter c to reflect the variability of the data set, and may have a ‘nugget’ discontinuity τ at the origin (see section 7.5.2.i).

It is worth noting that many of these models can be recreated or approximated by the more general Matern class of covariance functions (Wackernagel 1998, p336, Gneiting 2002a).

Although Whittle-Matern type models are very flexible, we chose not to implement them due to their computational expense, which must be undergone twice for every prediction location (once for the observation covariance matrix, once for the estimation covariance matrix; see section 7.9). Although this may seem limiting, significant flexibility can be achieved through the much simpler powered exponential model (Figure 7.1). Detailed notes on semivariogram specification can be found in section 7.11.2.v.

7.6.1.ii Nested models

The choice of permissible models may at first seem restrictive; however, additive combinations of permissible models always give rise to a permissible semivariance function. Hence, any number of semivariogram models can be combined in a linear additive manner to form complex nested models, which considerably increases the range of RF models possible. Although any number of models may be combined, the principle of parsimony is sensible when building a semivariogram model, as more

² **Note:** sometimes the exponential model is presented without the factor of 3 in the numerator (e.g. Cressie, 1990, pg. 61), in which case the less intuitive ‘integral range’ is solved for (Deutsch and Journal, 1998, pg. 25); all programs in the Edinburgh Space-Time Geostatistics package use the effective range convention in their calculations.

complex models do not necessarily improve results, and increase computation time. We recommend that the model should be as simple as possible, and generally comprise of less than three functions, which should preferably relate to the physical characteristics of the variable of interest.

7.6.1.iii Accounting for anisotropy

It is sometimes the case that the range of autocorrelation in a data set varies with the direction of the data pairs under consideration. This situation is referred to as *geometric anisotropy*, and is often observed when the underlying physical processes involved display directionality (e.g. down-wind dispersal of a Gaussian plume). Similarly it is possible to find a variation in the sill value of the semivariogram with direction; a situation referred to as *zonal anisotropy*. Whilst zonal anisotropy is not impossible *per se*, it is rarely encountered in practice, and often is apparent in cases where the sample space does not adequately cover the range of variation in all directions (Isobel Clark, *personal communication*). Zonal anisotropy can be incorporated by setting a very large range on the major axis of variation, such that the effect of the covariance structure is essentially nil perpendicular to the minor axis.

Anisotropy is dealt with by deforming the coordinate system such that all ranges appear equal (Figure 7.3). This is achieved by an affine transformation of the separation vector h :

$$h' = h + h|\sin(\theta)\psi|$$

where

$$\psi = \frac{\phi_{\min}}{\phi_{\max}} \quad (7.13)$$

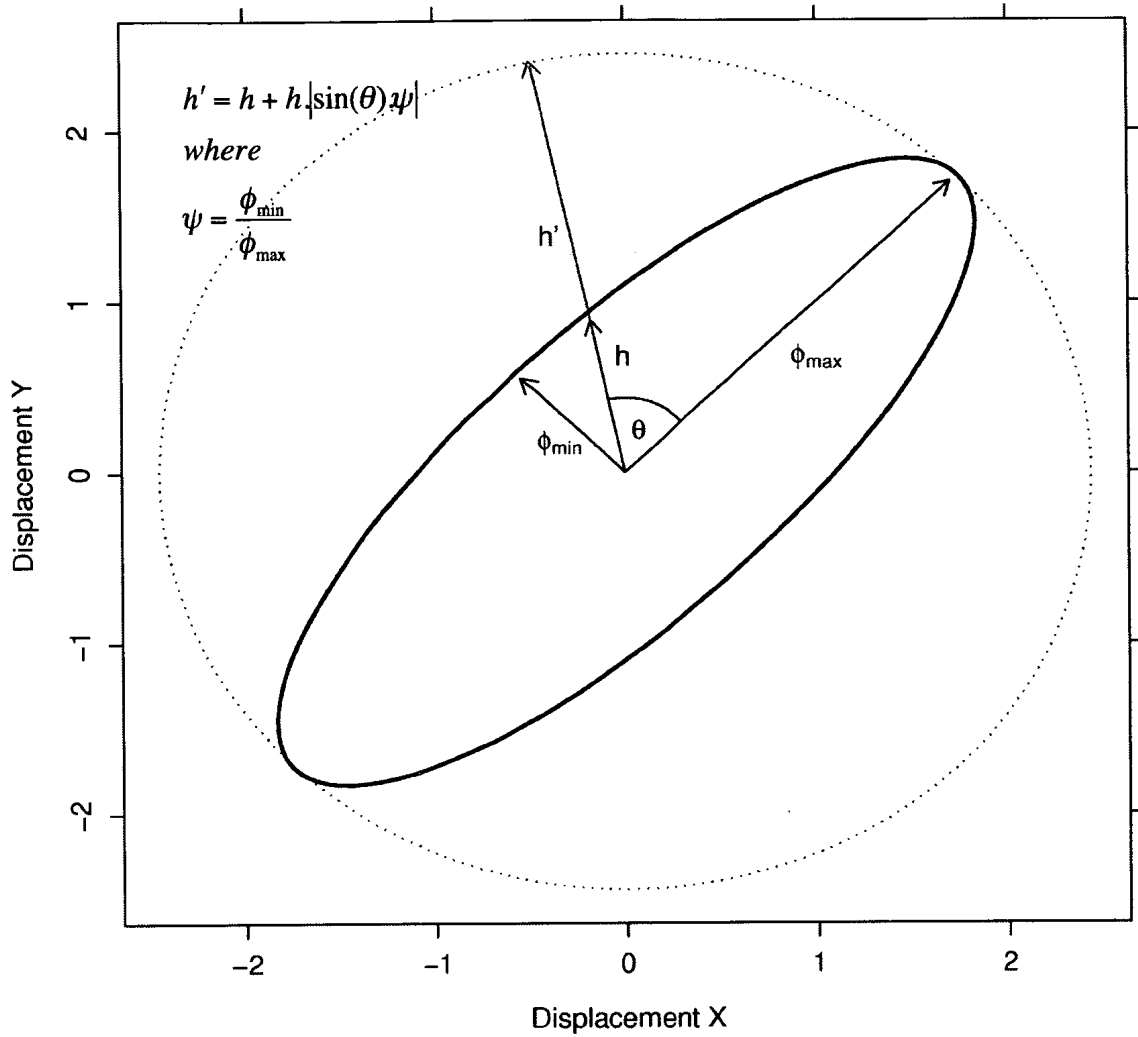


Figure 7.3 Affine transformation of anisotropy ellipse to isotropic coordinate system h' . The minor axis of variation is stretched until it is equivalent to the major axis using equation 4.8. The degree of stretching is dependant on the angle θ subtended by the distance vector h and the major semi-axis of variation, and the eccentricity of the anisotropy ellipse ψ .

The value ψ is the ratio between the minimum range and the maximum range of the semivariogram, and describes the eccentricity of the anisotropy ellipsoid. θ describes the angular difference between the coordinate vector and the direction of the axis of maximum variation $\phi_{\max}$. The new distance in the transformed coordinate system h' is then used to construct the covariance arrays for estimation.

7.6.1.iv Model Fitting

The method by which these models should be fit is subject to some dispute (see Goovaerts 1992, Diggle et al. 2002 for contrasting views); one school of thought is to fit the model by eye, so that the model appears visually satisfactory. Others use automated fitting procedures, either by least squares (Cressie 1985a) or likelihood based procedures (Diggle et al. 2002).

Whilst least squares (LS) methods provide a fit which is generally visually pleasing, fits may be sensitive to outliers, and in general the sensitivity of the semivariogram to changes in the conditioning data (as assessed by random draws from simulated surfaces) indicates that excessive 'trust' in the data points may produce misleading results (Diggle et al. 2002). Fitting by maximum likelihood (ML) generally requires some notion of the data error distribution, and although such approaches allow balancing of model fit and data uncertainty, they are open to criticism for subjectivity in the choice of distribution parameters, functional form for the prior error model etc.

Kriging estimates can be fairly resilient with respect to changes in the semivariogram specification (Cressie and Zimmerman 1992), although the estimation (Kriging) variances are sensitive to such changes. As a rule of thumb, LS fitting approaches work well when the purpose of the study is estimation, whilst ML based methods are more appropriate when inferences on the parameters of the spatio-temporal distribution are of more interest. We leave decisions on model fitting procedures to the discretion of the user, but urge against 'black box' fitting methods and favour a more interactive approach.

7.7 Modelling a Space-Time Regionalization

So far we have only discussed modelling of semivariograms in terms of spatial or temporal coordinates. Combining the spatial and temporal autocorrelation structures to form a complete model for the regionalisation has been an active area of research in the past decade, and comprehensive reviews can be found in (Kyriakidis and Journel 1999, De Cesare et al. 2001a, Gneiting 2002b, Gneiting et al. 2005). Strategies for combining spatial and temporal autocorrelation structures (which are generally discussed in their covariance form) can be broadly divided into *separable* and *non-separable* classes of models.

The earliest attempts at producing spatio-temporal covariance functions made use of separable models, with somewhat simplistic assumptions about the nature of spatio-temporal variability; either combining spatial and temporal covariance in an additive or multiplicative manner. The separable construction is tantamount to ignoring spatio-temporal interactions, and stating that spatial and temporal covariance display complete independence (Kyriakidis and Journel 1999). Few observed processes behave in this manner, and considerable effort has been made in seeking alternative non-separable representations.

Development of non-separable covariance functions began with metric models (Dimitrakopoulos 1994), whereby spatial and temporal separation units were converted to some common metric, and standard three-dimensional zonal anisotropy techniques used to produce the regionalisation (e.g. as implemented in GSLIB: Deutsch and Journel 1998). The attractive simplicity of this approach is somewhat offset by difficulties in specifying a common metric, and the loss of intuitive units to describe autocorrelation.

Subsequent to this, Cressie and Huang developed a set of permissible non-separable space-time covariance functions through Fourier inversion of one-dimensional covariance functions (Cressie and Huang 1999). Gneiting developed this approach to a Fourier free representation (Gneiting 2002b). These developments were an important step forward in spatio-temporal geostatistics, but it was not until the contribution of De Iaco et al. (2001) that these forms of stationary, non-separable covariance functions became generalized and straightforward to implement (see De Cesare et al. 2001a, b).

The product-sum covariance model of De Iaco (2001) allows the linear combination of arbitrarily complex covariance structures (including zonal and geometric anisotropy) in space and time, with full interaction. The product-sum representation incorporates the Cressie-Huang family of covariance functions and provides new, non-integrable forms that cannot be obtained through the Cressie-Huang representation (De Iaco et al. 2001). Due to the simplicity and ease of use of the product-sum representation, it is the representation of choice for non-separable covariance specification in all programs in the Edinburgh Space-Time Geostatistics package.

Non-stationary space-time covariance structures have been discussed (Kyriakidis and Journel 1999, 2001a), and generally rely on treatment of semivariogram parameters themselves as Gaussian RFs for the region of interest. Kyriakidis and Journel provide an interesting example of this hierarchical RF implementation for European pollution data (Kyriakidis and Journel 2001b). Although this implementation is powerful, it requires multiple regionalisations in order to build the nonstationary RF model, and requires a considerable investment of time in terms of semivariogram modelling and computational load. If such models are required, initial regionalisations may be

undertaken with the existing software, whilst final implementation is possible with minimal modification to the code.

In the discussion of models spatio-temporal covariance structures the following notational conventions will be used: Whenever referring to a property concerning patterns of spatial covariance, a subscript u will be applied. For properties of the covariance structure concerning time, a subscript t is applied. As in previous sections, h denotes separation distance, whilst $\hat{\gamma}$ indicates the semivariance, C denotes covariance, and the definitions of the semivariogram parameters (ϕ and τ) remain the same (see section 7.6.1). We denote the spatiotemporal semivariogram $\hat{\gamma}_{u,t}$, and refer to subsets of $\hat{\gamma}_{u,t}$ by indicating the range of separation distaces parenthetically, for example $\hat{\gamma}_{u,t}(h_u, h_t = 0)$ is the subset where all temporal separations are zero, i.e. only the spatial element of $\hat{\gamma}_{u,t}$.

7.7.1 Separable Space-Time Covariance: The Product Model

The simplest way to arrive at a permissible space-time covariance function is the product model, which is simply a scaled product of the separate spatial and temporal covariance models. An example of a product semivariogram model is provided in Figure 7.4. Fitting of this model proceeds as follows:

- First, only the simple spatial and temporal semivariograms are considered ($\hat{\gamma}_{u,t}(h_u, h_t = 0)$ and $\hat{\gamma}_{u,t}(h_u = 0, h_t)$ respectively), where h_u and h_t are the spatial and temporal separations. Valid semivariogram models must be fit to them (see section 7.6.1), estimating the spatial and temporal ‘partial’ ranges (ϕ_u, ϕ_t) and sills ($sill_u, sill_t$), and adding a ‘nugget’ discontinuity (τ_u, τ_t) at the origin to reflect spatial uncertainty if required.
- Having described the spatial and temporal behaviour separately, we examine the values of the semivariogram beyond the spatial and temporal ranges ($\hat{\gamma}_{u,t}(h_u > \phi_u, h_t > \phi_t)$) to find the global sill ($sill_g$).
- Calculate the weighting parameter k :

$$k = \frac{sill_g}{sill_u \cdot sill_t} \quad (7.14)$$

- The full covariance model is then arrived at as follows:

$$\begin{aligned} C_{u,t}(h_u, h_t) &= k \cdot C_u(h_u) \cdot C_t(h_t) \\ \text{where} \\ C_u(h_u) &= sill_u - \hat{\gamma}_{u,t}(h_u, h_t = 0) \\ C_t(h_t) &= sill_t - \hat{\gamma}_{u,t}(h_u = 0, h_t) \end{aligned} \quad (7.15)$$

7.7.2 Non-Separable Space-Time Covariance: The Product-Sum Model

In order to fit a product-sum covariance model, we proceed as described by De Iaco et al (2001):

- First, only the simple spatial and temporal semivariograms are considered ($\hat{\gamma}_{u,t}(h_u, h_t = 0)$ and $\hat{\gamma}_{u,t}(h_u = 0, h_t)$ respectively), where h_u and h_t are the spatial and temporal separations. Valid semivariogram models must be fit to them (see section 7.6.1), estimating the spatial and temporal ‘partial’ ranges (ϕ_u, ϕ_t) and sills ($sill_u, sill_t$), and adding a ‘nugget’ discontinuity (τ_u, τ_t) at the origin to reflect spatial uncertainty if required.
- Having described the spatial and temporal behaviour separately, we examine the values of the semivariogram beyond the spatial and temporal ranges ($\hat{\gamma}_{u,t}(h_u > \phi_u, h_t > \phi_t)$) to find the global sill ($sill_g$).
- We then check the validity of the fitted model, using the values of $sill_u$, $sill_t$ and $sill_g$ via the diagnostics detailed in De Cesare et al (2001), to ensure the resulting space-time semivariance function is conditionally negative definite. We calculate three diagnostic values k_1 , k_2 and k_3 as follows:

$$\begin{aligned}
 k_1 &= \frac{sill_u + sill_t - sill_g}{sill_u \cdot sill_t} \\
 k_2 &= \frac{sill_g - sill_t}{sill_u} \\
 k_3 &= \frac{sill_g - sill_u}{sill_t}
 \end{aligned} \tag{7.16}$$

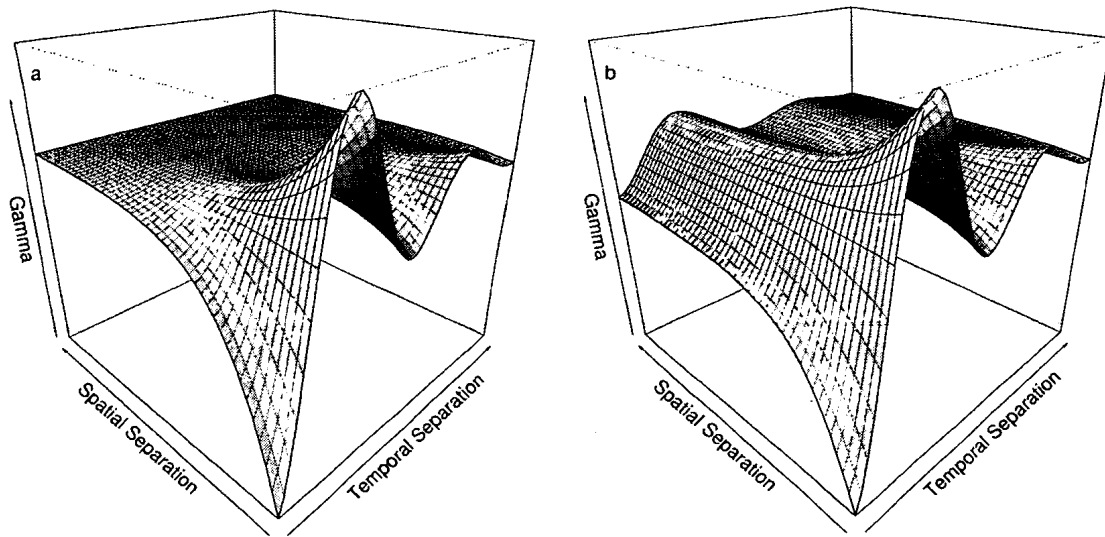


Figure 7.4 Comparison of spatiotemporal semivariogram representations: (a) product model (7.7.1). (b) product-sum model (7.7.2). Note the increased interaction in the product-sum model. Both plots share the same structures and parameters, and differ only in the spatiotemporal combination method.

- To ensure conditional negative definiteness of the resultant semivariogram is necessary that $k_1 > 0$, $k_2 \geq 0$, and $k_3 \geq 0$. If the above diagnostic constraints are met, the RF model is permissible and may be used for estimation/simulation purposes.
- The full covariance model is then arrived at as follows:

$$C_{u,t}(h_u, h_t) = k_1.C_u(h_u).C_t(h_t) + k_2.C_u(h_u) + k_3.C_t(h_t)$$

where

$$C_u(h_u) = sill_u - \hat{\gamma}_{u,t}(h_u, h_t = 0)$$

$$C_t(h_t) = sill_t - \hat{\gamma}_{u,t}(h_u = 0, h_t)$$
(7.17)

A comparison of the product-sum and product representations is provided in Figure 7.4.

7.8 Spatio-Temporal Estimation

All of the interpolation methods implemented in the Edinburgh Space-Time Geostatistics package produce estimates through weighted linear combination of a subset of the data $Z = \{z_1(u, t) \dots z_n(u, t)\}$, selected on the basis of spatial and temporal distance from the estimation datum. Spatial effects are decomposed into a global trend m , and a high frequency autocorrelated residual component, formed from a weighted linear combination of the residuals. Therefore the only difference in the prediction algorithms is the method by which the weights (λ) are derived:

$$z^*(u, t) = m(u, t) + \sum_{i=1}^n \lambda_i(u, t) [z_i(u, t) - m(u, t)] \quad (7.18)$$

7.8.1 Inverse Distance Weighting

Inverse distance weighted averages produce estimates of the variable of interest by linear combination of the observations, such that data points closer to the estimation location are ascribed more prominence than those further away. Weights are ascribed to a subset of the total data pool, such that the conditioning data follow a spatial power law. The weights are rescaled such that they sum to one, preventing the estimation exceeding the range of the conditioning data. Any power ω can be used, but most commonly an inverse squared power law is encountered in the manner of Newton's Gravity model; hence the alternative moniker 'Spatial Gravity Models'. Estimates are calculated as follows:

$$z^*(u, t) = \sum_{i=1}^n w_i(u, t) \cdot z_i(u, t)$$

where

$$w_i(u, t) = \frac{1/d_i^w}{\sum_{i=1}^n 1/d_i^w} \quad (7.19)$$

Where w are the weights, and d_i is the separation distance between $z_i(u, t)$ and z^* .

If the estimation location is coincident with an observation, the estimate takes the value of the observation. In the case of global trends, residual data may be interpolated, and the trend added back in after interpolation.

Currently *IDW.exe* does not incorporate temporally adjacent data into the conditioning data; estimates are produced one time step at a time from the available spatial neighbours.

7.8.2 Kriging Methods

Kriging refers to a set of multiple linear regression procedures by which the best linear unbiased estimate of an unobserved datum value is arrived at by the weighted linear combination of surrounding observations, such that the prediction error is minimized. A good introductory text is provided by Isaacs and Srivastava (1989); for a historical perspective, see Cressie (1990). The weights ascribed to each observation take into consideration the clustering of the data locations, and the proximity of each observation to the prediction location. These spatial effects are included via reference to the autocorrelation structure of the data set, as summarized by the semivariogram. The result of considering distances between the conditioning data is that points from over-sampled locations are down weighted.

Kriging is a regression procedure of the form $y = mx + c$. In standard notation, we write:

$$z^*(u, t) = \lambda(Z - m) + m \quad (7.20)$$

Where z^* is the value of an unobserved spatio-temporal location u, t to be estimated, λ is the vector weights, Z is the vector of observations and m is the mean. Typically we work with a sub-set of observations, say the $n = 12$ closest observations to $z^*(u, t)$. In this case, m is the local mean m^* . Thus $z^*(u, t)$ is predicted by the sum of the local mean, plus the spatially autocorrelated deviation from that mean. Differences between Kriging methods reside in the m^* term of equation 7.20. An excellent summary of all Kriging methods described is included in Goovaerts (1999).

7.8.2.i Simple Kriging

Simple Kriging is used in the situation where the mean of the variable of interest m is known across the whole study region. This mean need not be the same at all locations, but the assumption is that the sampling design is sufficient to ensure that calculation of the mean is not affected by data clustering. The mean is subtracted from the observations prior to estimation, and the residual values are used to correct the estimation surface, by the addition of spatially coloured noise. The local correction is a function of the data locations (clustering) and the distance between the conditioning data and the estimation datum, imposed through the Kriging weights λ . The weights are arrived at by solving the following system of linear equations:

$$z^*(u, t) = m(u, t) + \sum_{i=1}^n \lambda_i(u, t) \cdot [z(u_i, t_i) - m(u, t)]$$

$$\sum_{i=1}^n \lambda_i(u, t) \cdot C[z(u_i, t_i) - z(u_j, t_j)] = C[z^*(u, t) - z(u_i, t_i)] \quad \text{where } j = 1, \dots, n$$

(7.21)

Here $C[z(u_i, t_i) - z(u_j, t_j)]$ and $C[z^*(u, t) - z(u_i, t_i)]$ are the covariance between observations, and the covariance between observations and the prediction location u, t respectively. These values are obtained by looking up the separation distance b against the semivariogram model. In the case of an underlying trend across the region of interest, the m can take the value of the deterministic trend, which is calculated in advance for each estimation datum.

7.8.2.ii Ordinary Kriging

In the case that the mean value is known in advance, we proceed by Simple Kriging (SK), and m takes the known value of the mean. However, more often the mean is unknown, or is not easily calculated in advance due to data clustering: Estimation then proceeds by ordinary Kriging (OK), where the unknown mean m must be estimated simultaneously with the autocorrelated residual component.

Usually we condition on a subset of n available data, and thus $m^*(u, t)$ represents the local mean of the variable of interest. Given that we only require constant mean and variance within the neighbourhood of the selected data, such moving window approaches allow some degree of robustness to the assumption of stationarity. The OK estimate is thus arrived at by solving the following system of equations:

$$\begin{aligned}
 z^*(u, t) &= \sum_{i=1}^n \lambda_i(u, t) \cdot z(u_i, t_i) \\
 \begin{cases} \sum_{i=1}^n \lambda_i(u, t) \cdot C[z(u_i, t_i) - z(u_j, t_j)] + \mu = C[z^*(u, t) - z(u_j, t_j)] \\ \sum_{i=1}^n \lambda_i(u, t) = 1 \end{cases} & \quad \text{where } j = 1, \dots, n
 \end{aligned}
 \tag{7.22}$$

As before $C[z(u_p, t_p) - z(u_p, t_j)]$ and $C[z^*(u, t) - z(u_p, t_j)]$ are the covariance between observations, and the covariance between observations and the prediction location u, t respectively. These values are obtained by looking up the separation distance h against the semivariogram model. The main difference between SK and OK is in the system of equations governing the calculation of λ . Equation system 7.21 is modified by the addition of a Lagrange multiplier μ , necessary to satisfy the un-biasness constraint that the weights sum to one.

7.8.2.iii Kriging With an External Drift

A more complex spatio-temporal regression model can be formulated by extending the framework established in equation 7.20 to include extra covariates in m^* . This leads to a family of Kriging systems referred to as Kriging with a trend (KT). Here m^* contains more terms than the intercept only solutions illustrated above: The m^* component of a KT system contains an intercept b_0 , and k slope parameters $b_1, \dots, b_k$. This generalisation allows fitting of linear, polynomial, or Fourier type basis functions combined in arbitrarily complex trend models. However, in practice k typically ≤ 5 , and trend models are restricted to low order polynomials (Deutsch and Journel 1998). This partitioning of the data into a large-scale trend component, and a stationary, spatially autocorrelated residual component is the rationale of non-stationary geostatistics:

$$z^*(u, t) = \lambda(Z - m^*) + m^*$$

where

(7.23)

$$m^* = b_0 + \sum_{k=1}^K b_k S_k$$

Where z^* is value of the unobserved datum location u to be estimated, λ is the vector weights, Z is the vector of observations, b_0 is the intercept, b_k are slope parameters and S_k are vectors of auxiliary variables recorded at all observation and prediction locations $S_k = \{s_1^k(u, t), \dots, s_n^k(u, t)\}$.

Universal Kriging (UK) is the simplest form of KT system, where m is a function of the coordinates, allowing large scale gradients to be dealt with by the Kriging system (i.e. non-stationary problems can be solved). The ordinary Kriging system can be seen as a special case of UK, where $k=0$ (Goovaerts, 1999).

Kriging with an external drift (KED) refers to the case where we use secondary covariates 'external' to the semivariance calculation for our data. Here we require that the variation of the secondary data be smoothly and linearly related to the variable of interest (Deutsch and Journel, 1998). The covariates must be sampled at all observation and all prediction locations.

Typically KED systems are restricted to a single covariate, although theoretically, any number of covariates may be included. The idea is that the extra covariates inform the interpolation, so that more easily available data, such as remote sensing observations, can be used to improve interpolation skill. The method may also be used to impose known physical trends on the interpolation, for example, the decrease in temperature with increased elevation (Hudson and Wackernagel 1994), or as a form

of model-data fusion, using model output as the external drift (e.g. Wackernagel 1998 p297).

Again, we usually condition on a subset of n data points, with $m^*(u,t)$ returning the local trend. Estimation proceeds by solving the following system of equations:

$$\begin{aligned}
 z^*(u,t) &= \sum_{i=1}^n \lambda_i(u,t) \cdot z(u_i, t_i) \\
 \left\{ \begin{aligned}
 &\sum_{i=1}^n \lambda_i(u,t) \cdot C[z(u_i, t_i) - z(u_j, t_j)] + \mu_1(u,t) + \mu_2(u,t) \cdot s(u_j, t_j) \\
 &= C[z^*(u,t) - z(u_i, t_i)] \\
 &\sum_{i=1}^n \lambda_i(u,t) = 1 \\
 &\sum_{i=1}^n \lambda_i(u,t) \cdot s(u_i, t_i) = s(u,t)
 \end{aligned} \right. \quad \text{where } j = 1, \dots, n
 \end{aligned}
 \tag{7.24}$$

Here we add a further constraint to equation system 7.22, requiring an extra Lagrange multiplier μ_2 . This ensures that the dot product of the weights (λ) and the vector of secondary variable (s) equals the observed value of the secondary variable at the prediction location. It is possible to retrieve local values of the intercept and slope parameters with simple modifications to equation system 7.24 (see section 7.9.1.iii). This is particularly useful in the model-data fusion case, where model efficiency can be assessed locally by tracking deviations from $b_0 = 0$ and $b_1 = 1$.

Addition of extra trend terms would require an additional Lagrange multiplier for each term, involving minor code modifications. Further details and examples can be found in Wackernagel (1998).

7.9 Practical Aspects of Kriging

7.9.1.i Calculating Kriging weights

For all Kriging algorithms implemented, the solution of the linear equations to derive λ follows the same general pattern. An illustration of the technique in matrix formulation highlights the similarities between the various algorithms.

In order to retrieve λ from the observed data values, we first produce two distance tables OO_b and OE_b . Matrix OO_b is the square n by n distance matrix for the conditioning data, whilst vector OE_b contains the distances between the conditioning data and the estimation datum:

$$OO_h = \begin{bmatrix} h(z_1, z_1) & h(z_2, z_1) & \dots & h(z_n, z_1) \\ h(z_1, z_2) & h(z_2, z_2) & \dots & h(z_n, z_2) \\ \vdots & \vdots & \ddots & \vdots \\ h(z_1, z_n) & \vdots & \vdots & h(z_n, z_n) \end{bmatrix} OE_h = \begin{bmatrix} h(z_1, z^*) \\ h(z_2, z^*) \\ \vdots \\ h(z_n, z^*) \end{bmatrix} \quad (7.25)$$

These distances are easily converted to covariances, by reference to the semivariogram model (see section 7.5.2.ii): Matrix OO is now the observation covariance matrix, and vector OE is the observation-estimation covariance vector. To calculate simple Kriging weights, we invert matrix OO , and multiply by vector OE :

$$\begin{bmatrix} \lambda_1 \\ \lambda_2 \\ \vdots \\ \lambda_n \end{bmatrix} = \begin{bmatrix} C(z_1, z_1) & C(z_2, z_1) & \dots & C(z_n, z_1) \\ C(z_1, z_2) & C(z_2, z_2) & \dots & C(z_n, z_2) \\ \vdots & \vdots & \ddots & \vdots \\ C(z_1, z_n) & \vdots & \vdots & C(z_n, z_n) \end{bmatrix}^{-1} \times \begin{bmatrix} C(z_1, z^*) \\ C(z_2, z^*) \\ \vdots \\ C(z_n, z^*) \end{bmatrix} \quad (7.26)$$

In the case of SK weights, the vector $\Lambda = \{\lambda_1 \dots \lambda_n\}$ sums to zero. However, in the case of OK or KED, we require that the weights sum to one. The problem is one of

minimisation with a constraint, which can be achieved by *via* the use of Lagrange multipliers. The Lagrange multipliers allow us to solve for m simultaneously with the solution for the weights.

For each extra term in m , we require a Lagrange multiplier $M = \{\mu_1 \dots \mu_k\}$, where k is the number of model terms. In the case of simple Kriging $k=0$, whilst for ordinary Kriging $k=1$: We solve for the intercept only (null model) by adding an extra row and column to OO and OE in which all values equal one, except the bottom right element of OO which contains a zero. An additional row on the weights vector then accommodates the Lagrange multiplier:

$$\begin{bmatrix} \lambda_1 \\ \lambda_2 \\ . \\ \lambda_n \\ \mu \end{bmatrix} = \begin{bmatrix} C(z_1, z_1) & C(z_2, z_1) & . & C(z_n, z_1) & 1 \\ C(z_1, z_2) & C(z_2, z_2) & . & C(z_n, z_2) & 1 \\ . & . & . & . & . \\ C(z_1, z_n) & C(z_2, z_n) & . & C(z_n, z_n) & 1 \\ 1 & 1 & 1 & . & 0 \end{bmatrix}^{-1} \times \begin{bmatrix} C(z_1, z^*) \\ C(z_2, z^*) \\ . \\ C(z_n, z^*) \\ 1 \end{bmatrix} \quad (7.27)$$

For more complex regression models $k>1$, and we implement the KED algorithm. Addition of terms follows the same pattern as above, augmenting OO and OE with extra rows and columns for the additional Lagrange multipliers. Retrieving λ for an arbitrarily complex regression model KED is then:

$$\begin{bmatrix} \lambda_1 \\ \lambda_2 \\ . \\ \lambda_n \\ \mu_1 \\ . \\ \mu_k \end{bmatrix} = \begin{bmatrix} C(z_1, z_1) & C(z_2, z_1) & . & C(z_n, z_1) & s_1^1 & . & s_1^k \\ C(z_1, z_2) & C(z_2, z_2) & . & C(z_n, z_2) & s_2^1 & . & s_2^k \\ . & . & . & . & . & . & . \\ C(z_1, z_n) & C(z_2, z_n) & . & C(z_n, z_n) & s_n^1 & . & s_n^k \\ s_1^1 & s_2^1 & . & s_n^1 & 0 & . & 0 \\ . & . & . & . & . & . & . \\ s_1^k & s_2^k & . & s_n^k & 0 & . & 0 \end{bmatrix}^{-1} \times \begin{bmatrix} C(z_1, z^*) \\ C(z_2, z^*) \\ . \\ C(z_n, z^*) \\ s_*^1 \\ . \\ s_*^k \end{bmatrix} \quad (7.28)$$

Where k is typically less than five. S' is normally a vector of ones, to solve for the intercept, whilst S^2 to S^k are vectors of secondary variables, which are linearly related to the variable of interest (see section 7.8.2.iii). Again, the bottom right section of OO is filled with zeros.

Sometimes a multiple intercept model is desirable, e.g. to impose different means by vegetation or soil type. In this case, vectors S' to S^k contain dummy variables, with binary coding for the factor levels. Mixed effects models are readily incorporated by combination of the above techniques. The only limit to the complexity of the model is the need to invert matrix OO , a time consuming and not always stable process: As the complexity of the model increases, the likelihood of producing a singular (degenerate) matrix OO increases. Again, the principle of parsimony is a good guiding rule.

Currently, the programs in the Edinburgh Space-Time Geostatistics package only allow for simple linear models with $k < 2$. However, more complex models are possible with minor alteration to the code.

7.9.1.ii Kriging Variances

One of the major attractions of the Geostatistical method is the ability to retrieve estimation variances. Kriging variances are easily obtained by subtracting the dot product of the observation-estimation covariance vector OE and the weights vector Λ from the global sill ($C_{0,0}$). In the case of OK or KED, the full augmented vectors are used, so the resulting variance takes into account the uncertainty associated with the trend model m :

$$\sigma^2(u, t) = C_{0,0} - \begin{bmatrix} \lambda_1 \\ \lambda_2 \\ \vdots \\ \lambda_n \\ \mu_1 \\ \vdots \\ \mu_k \end{bmatrix}^T \times \begin{bmatrix} C(z_1, z^*) \\ C(z_2, z^*) \\ \vdots \\ C(z_n, z^*) \\ s_*^1 \\ \vdots \\ s_*^k \end{bmatrix} = C_{0,0} - \sum_{i=1}^n \lambda_i C(z_i, z^*) - \sum_{j=1}^k \mu_j s_*^j \quad (7.29)$$

7.9.1.iii Parameter Retrieval and Filtering

In the case of OK and KED, it is often desirable to retrieve the local parameters $B^* = \{b_0^* \dots b_k^*\}$ from equation 7.23. In the case of OK, we may be interested in declustered local mean of the n conditioning data; in this case we must filter off the high frequency autocorrelated noise ε to obtain the value m . This is easily achieved by modification of the Kriging weights (equation 7.27) to ignore the effect of the conditioning data. We filter ε by setting the first n elements of OE to zero:

$$\begin{bmatrix} \lambda_1 \\ \lambda_2 \\ \vdots \\ \lambda_n \\ \mu \end{bmatrix} = \begin{bmatrix} C(z_1, z_1) & C(z_2, z_1) & \vdots & C(z_n, z_1) & 1 \\ C(z_1, z_2) & C(z_2, z_2) & \vdots & C(z_n, z_2) & 1 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ C(z_1, z_n) & C(z_2, z_n) & \vdots & C(z_n, z_n) & 1 \\ 1 & 1 & \vdots & 1 & 0 \end{bmatrix}^{-1} \times \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix} \quad (7.30)$$

The value of m is then Kriged with the modified vector Λ .

In the case of KED, the value m is retrieved as above, by setting elements $OE_{1:n}$ to zero, and $OE_{n+1:k}$ to ones. Individual KED parameters $B^* = \{b_i^* \dots b_k^*\}$ can be retrieved by substituting OE for the Kronecker delta function $(\delta_{n+i,j})$ centred on the i^{th}

trend parameter, which is one for OE'_{n+i} and zero otherwise; i.e. OE' contains a binary coding for the parameter we wish to retrieve:

$$OE' = \delta_{n+i,j} \begin{cases} 1 & \text{if } j = n+i \\ 0 & \text{if } j \neq n+i \end{cases} \quad \text{where } j = 1 \dots n+k \quad (7.31)$$

7.9.1.iv Exactitude of Kriging

In general, discussions of Kriging specify the nugget variance at $h=0$ as zero, and τ elsewhere (Goovaerts 1997). This formulation ensures that the Kriging estimate for a location at which a conditioning observation exists is exactly equal to the observed value. In this sense, Kriging is referred to as an 'exact interpolator'. This is desirable if we believe our data is infallible, and assumes the primacy of the data over the specified RF model. This approach leads to very 'spiky' surfaces where observations are collocated with grid nodes, due to the discontinuity in the semivariogram model.

In reality, we are faced with imperfect data and an imperfect model representation; whilst the observations correct the local model m in the kriging estimate, we may also wish m to correct the observations when z^* is collocated with a conditioning datum. If we allow the nugget variance to equal τ for all h we no longer strictly honour the observations, and the estimate z^* at an observation location is equivalent to a spatio-temporal assimilation of the observations with m . In this case the estimate at an observation location will be heavily weighted towards the observed value, but the estimate will be drawn towards a value concurrent with the surrounding observations (subject to the semivariogram) and any additional external variables included in the trend model. The degree to which surrounding observations and m correct the data is specified by the magnitude of τ .

It is our contention that the model data fusion approach resultant from allowing a non-zero nugget effect when $b=0$ confers a considerable advantage for the poorly or pseudo-replicated data sets generally used in upscaling studies, and from which the intrinsic stationarity assumption arises (see section 7.4). As such, all semivariance models in the Edinburgh Space-Time Geostatistics package are implemented with a nugget effect equal to τ for all separation distances.

7.10 Assessing Spatio-Temporal Uncertainty

When simple estimation is the goal of a study, evaluation of the RF at a set of locations via Kriging provides an unbiased estimate of the variable of interest, and a Kriging variance, reflecting the uncertainty associated with the draw from the RF. The Kriging variance reflects our prior conception of the data; that it is normally distributed, intrinsically stationary, and autocorrelated with a known and accurately described covariance structure, as inferred from the semivariogram.

Although values derived from Kriging provide optimal estimates in the least squares sense, the results tend to be smoothed, losing the extremes of the data distribution. This smoothing is non-uniform, and occurs mainly at locations separated by large distances from the conditioning data; thus local variability appears (non-intuitively) greater where more observations are present (Goovaerts 1997). Furthermore, the semivariogram is only honoured if we compare observation locations with estimation locations, although short scale variation may be smoothed by the Kriging algorithm: In general Kriging estimates do not reproduce the semivariogram globally, although these effects may be mediated to some degree by *post-hoc* processing (Olea and Pawlowsky 1996).

Whilst Kriging variances appropriately describe model uncertainty, they do not directly relate to error in the sense of model data divergence. In fact, the Kriging variances are arrived at by reference only to the spatio-temporal arrangement of data values, and ignore the data values entirely. In this sense the Kriging variance is essentially a ranking score of the data configuration (Journel and Rossi 1989). This is not entirely satisfactory if the goal of the study is to make inferences on the errors associated with estimation of the variable of interest, e.g. when the resulting estimated fields are to be used as inputs to a model; a common application of upscaled surfaces. In this case, a set of random draws from the spatio-temporal RF *via* simulation is more appropriate. One such method of drawing realisations from the RF is *Sequential Gaussian Simulation*. Discussions of alternatives can be found in (Deutsch and Journel 1998).

7.10.1 Sequential Gaussian Simulation

Sequential Gaussian simulation can be employed in order to produce estimates of local error which reflect the data values, and globally preserve the ‘texture’ of the data variability. The procedure is very similar to the Kriging methods outlined above, with one critical difference: Upon estimation, a draw from the RV $z^*(u, t)$ is added to the data heap used to condition subsequent estimates. Thus as we step through the estimation locations, the size of the conditioning data pool increases, and each estimate is conditionally dependant on all other values in the estimation field. The algorithm functions as follows:

1. Initialise a random visiting schedule for the grid of G locations, with a data heap of n observations $Z = \{z_1(u, t) \dots z_n(u, t)\}$. Initialise with $i=1$.
2. Visit the i^{th} node of the grid and estimate the expectation and variance *via* Kriging conditioned on the values in the data heap.

3. Draw a random value from the Gaussian distribution of the node, defined by the Kriging estimate (mean) and Kriging variance. The resultant value was the SGS estimate z_i^* .
4. The realization z_i^* was then treated as an observation for subsequent estimates, and added to the data heap ($n+i$ conditioning data).
5. Iterate from 2 until all grid locations were visited ($i=G$).

As in all geostatistical techniques, it is possible to incorporate covariates into the simulations. Randomisation of the visiting schedule ensures each field simulated from the RF will be a unique realisation of the spatio-temporal model, whilst conditioning on all values in the heap ensures reproduction the semivariogram globally.

Sequential Gaussian simulation is available in the Edinburgh Space-Time Geostatistics package using the *Gaussim.exe* program.

7.11 Program Notes and Instructions

7.11.1 IDW.exe

7.11.1.i Data format

All input and output to *IDW.exe* is handled via tab delimited text files. Data are supplied with a strict format: The first line contains a heading in inverted commas. The second line contains, seven column names, and subsequent lines contain the sequence of observations. Missing data flags are not supported. Each data table entry must contain a unique observation ID, a station ID, a sequence of three spatio-temporal coordinates, an observation value and a secondary mean/trend value. Observation Ids

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(OID) identify each unique datum, whilst station IDs (SID) identify the time series to which each observation belongs:

"3164 January temperature observations from 112 sites across Oregon, USA"							
OID	SID	EAST	NORTH	TIME	TEMP.C	MEAN.C	
1	1	633717.8	4878503	1	7.778	3.429	
2	1	633717.8	4878503	2	6.110	1.836	
3	1	633717.8	4878503	3	6.672	3.319	
...							
3163	112	614058.7	4921431	30	-4.693	1.940	
3164	112	614058.7	4921431	31	6.679	4.661	

7.11.1.ii Notes

The mean is removed from the observations by the program prior to interpolation then added to the estimates before output. As such, the observation values should be raw.

7.11.1.iii Grid file

IDW.exe requires a list of the estimations coordinates for interpolation. Coordinates should be supplied as a tab delimited text file containing a header followed by a list of spatiotemporal coordinates and a mean function. The file must be in the following format:

EAST	NORTH	TIME	MEAN
562618	4903722	1	3.429
563618	4903722	1	3.429
564618	4903722	1	3.429
...			
661618	5003722	31	4.661
662618	5003722	31	4.661

The secondary data may contain the (local) mean or the values of a trend model.

7.11.1.iv Parameter file

IDW.exe is parameterised with a tab delimited text file laid out in strict format:

```

FILENAMES [Data,Grid,Output]:
~/ED_ST_Geostats/Data/Data.txt
~/ED_ST_Geostats/Data/Grid.txt
~/ED_ST_Geostats/Outputs/IDW_out.txt

WEIGHTING FUNCTION:
2

MODE [0=Debug,1=Default,2=Jack-knife]:
1

SEARCH STRATEGY [Neighbours]:
16

```

- Lines 2:4 contain the path and filename of the data file, grid file and output file respectively.
- Line 7 contains the power to use for inverse distance weighting
- Line 10 contains a mode switch, selecting between debug, default or jack-knife modes:
- Debug: Returns a screen prompt containing information on the conditioning data and interpolation result for each point in the grid file. This is a good way to experiment with parameterisation of the program. Results are stored in the output file.
- Default: Interpolates each point in the grid file in turn with minimal output to screen. Results are stored in the output file.
- Jack-knife: Temporarily excludes one data point at a time from the observations, and estimates its value from the remaining data set. This is a good way to assess the interpolation skill given the selected parameters. Results are stored in a modified output file.

- Line 13 contains the number of conditioning data to use in the estimation: the n data points closest to the estimation grid coordinate are selected.

7.11.1.v *Output file*

IDW.exe returns a tab delimited text file containing various pieces of information on the interpolation, depending whether the program was run in debug/default or Jack-knife mode. Standard output contains the following columns:

- Columns 1:3 contain the spatio-temporal coordinates of the datum.
- Column 4: The IDW estimate.
- Column 5: The supplied trend value.
- Column 6: The interpolated residual value (estimate minus the trend).
- Column 7: The nearest neighbour distance.
- Column 8: The mean distance of the conditioning data.
- Column 9: The SID of the nearest conditioning datum.

When running in Jack-knife mode, the following columns are returned:

- Column 1: The SID of the Jack-knifed observation.
- Columns 2:4 contain the spatio-temporal coordinates of the datum.
- Column 5: The IDW estimate.
- Column 6: The supplied trend value.
- Column 7: The interpolated residual value (estimate minus the trend).
- Column 8: The observed value.
- Column 9: The error (observed minus estimated).
- Column 10: The nearest neighbour distance.
- Column 11: The mean distance of the conditioning data.

- Column 12: The SID of the nearest conditioning datum.

7.11.2 Geostats.exe

7.11.2.i Data format

All input and output to *Geostats.exe* is handled via tab delimited text files. Data are supplied with a strict format: The first line contains a heading in inverted commas. The second line contains, seven column names, and subsequent lines contain the sequence of observations. Missing data flags are not supported. Each data table entry must contain a unique observation ID, a station ID, a sequence of three spatio-temporal coordinates, an observation value and a secondary value. Observation Ids (OID) identify each unique datum, whilst station IDs (SID) identify the time series to which each observation belongs:

"3164 January temperature observations from 112 sites across Oregon, USA"							
OID	SID	EAST	NORTH	TIME	TEMP.C	MEAN.C	
1	1	633717.8	4878503	1	7.778	3.429	
2	1	633717.8	4878503	2	6.110	1.836	
3	1	633717.8	4878503	3	6.672	3.319	
...							
3163	112	614058.7	4921431	30	-4.693	1.940	
3164	112	614058.7	4921431	31	6.679	4.661	

7.11.2.ii Notes

In the case of SK the mean is removed from the observations by the program prior to interpolation, then added to the estimates before output. As such, the observation values should be raw, and the secondary value should contain the (local) mean.

SK can be used to incorporate complex trend models or remotely sensed products with the data; in this case the secondary data values should contain model

predictions for each datum. Again, raw observations of the primary variable should be supplied.

In the case of OK, the secondary data value is redundant. However, the program will encounter an error if the column is not filled; a reference value may be included, or zero padding employed.

If global trends are present and the user wishes to interpolate the residuals with OK, trends should be removed from the data prior to analysis with *Geostats.exe* and added into the estimates manually.

In the case of KED, the secondary value can be any auxiliary data linearly related to the primary variable. Again, raw observations of the primary variable should be supplied.

7.11.2.iii Grid file

The *Geostats.exe* grid file is identical in format to that of *IDW.exe*. However, in the case of *Geostats.exe*, the purpose of the secondary data changes with the Kriging method selected.

In the case of SK, the secondary data values are used in an identical fashion to *IDW.exe*, i.e. the interpolated residuals are added to the grid secondary data value to form the estimate.

In the case of OK, the secondary data value is redundant. However, the program will encounter an error if the column is not filled; a reference value may be included, or zero padding employed.

In the case of KED, the secondary data value contains the auxiliary variable, which must be known at all observation and grid locations.

7.11.2.iv Parameter file

Parameters are supplied to *Geostats.exe* in a tab delimited text file with strict format:

```

FILENAMES [Data,Grid,Output]:
~/ED_ST_Geostats/Data/Data.txt
~/ED_ST_Geostats/Data/Grid.txt
~/ED_ST_Geostats/Outputs/KED_out.txt

METHOD [0=SK,1=OK,2=KED]
2

MODE [0=Debug,1=Default,2=Jack-knife]:
1

SEARCH STRATEGY [Neighbours,Window]:
32      7

STRUCTURES [Spatial,Temporal]:
2      1

TAU [Spatial,Temporal]:
0.02    0.0

GLOBAL SILL:
32.46

COVARIANCE MODEL [0=Product,1=Product-Sum]:
1

Spatial      ]----- Theta -----[ Rotation
=====
Spherical      3.01      9.67      9.67      0.0      0.0
Exponential    6.99     196.55     196.55     0.0      0.0

Temporal      ]----- Theta -----[
=====
Exponential    23.13      6.6      0.0

```

- Lines 3:4 specify the data file, grid file and output file names respectively
- Line 7 contains a mode switch to select between the Simple Kriging (SK), Ordinary Kriging (OK) and Kriging with an External Drift (KED) algorithms.

- Line 10 contains a mode switch, selecting between debug, default or jack-knife modes:
 - Debug: Returns a screen prompt containing information on the conditioning data and interpolation result for each point in the grid file. This is a good way to experiment with parameterisation of the program. Results are stored in the output file.
 - Default: Interpolates each point in the grid file in turn with minimal output to screen. Results are stored in the output file.
 - Jack-knife: Temporarily excludes one temporal vector at a time from the observations, and estimates its values from the remaining data set. This is a good way to assess the interpolation skill given the selected parameters. Results are stored in a modified output file.
- Line 13 parameterises the search strategy: it should consist of two integer values enumerating the number of data values to select from each time step, and the size of the temporal window to use.
 - Neighbours specifies the number of values to select from each window increment; the total number of conditioning data is therefore $neighbours + (neighbours * window)$.
 - Window specifies the number of timesteps to include in the search. The strategy is $\pm window$, so a value of 7 will incorporate data from a one fortnight window.
- Line 16 contains two integer values specifying the number of semivariogram structures (not including the nugget) to implement in space and time respectively.

- Line 19 contains the nugget variances for the spatial and temporal semivariograms respectively.
- Line 22 contains the global sill.
- Line 25 is a binary switch to select between the separable product, or non-separable product-sum covariance models.
- Lines 29 onwards specify the semivariogram models in space and time.

7.11.2.v *Semivariogram specification*

- ***Spatial model specifictaion***

Each line of the semivariogram specification should contain six elements: a structure name, a vector of four parameters, and a rotation parameter.

- Structure name: Linear, Power, Exponential, Spherical, Gaussian, Quadratic, Hole or Dampened Hole (see section 7.6.1).
- Theta1 specifies the contribution of the structure, and can take any non-zero real value.
- Theta2 and Theta3 are the ranges of the minor and major semi-axes of the anisotropy ellipse respectively. In the case of the linear and power models, these values are redundant, and can be zero padded.
- Theta4 is an auxiliary parameter required by some models, taking on the following values:
 - The power value ω for the powered exponential and power models.
 - The dampening length for the dampened hole model.
- Rotation: A rotation parameter to specify the direction of the major axis of variation in degrees from north (north = 0).

- ***Temporal Model Specification***

Each line of the temporal model specification should contain a list of four elements: A structure name, and a sequence of three parameters.

- Structure name, as above.
- Theta1 is the contribution of each structure of the model.
- Theta2 is the range of the structure.
- Theta3 contains auxiliary parameters, required by some models: See above.

7.11.2.vi *Output file*

Geostats.exe returns a tab delimited text file containing various pieces of information on the interpolation, depending whether the program was run in debug/default or Jack-knife mode. Standard output contains the following columns:

- Columns 1:2 contain the spatial coordinates of the datum.
- Column 3: The supplied secondary value.
- Column 4: The temporal coordinate.
- Column 5: The local mean m .
- Column 6: The Kriging estimate.
- Columns 7:8 contain the Kriging variance and standard deviation respectively.
- Column 9:10 contain the retrieved intercept and slope parameters when using KED.
- Column 11: The nearest neighbour distance.
- Column 12: The mean distance of the conditioning data.
- Column 13: The mean secondary value of the conditioning data.

- Column 14: The SID of the nearest conditioning datum.

When running in Jack-knife mode, the following columns are returned:

- Column 1: The SID of the Jack-knifed temporal vector.
- Columns 2:3 contain the spatial coordinates of the datum.
- Column 4: The supplied secondary value.
- Column 5: The temporal coordinate.
- Column 6: The observed value.
- Column 7: The local mean m .
- Column 8: The Kriging estimate.
- Columns 9:10 contain the Kriging variance and standard deviation respectively.
- Column 11:12 contain the retrieved intercept and slope parameters when using KED.
- Column 13: The Jack-knife error (observed value – Kriging estimate).
- Column 14: The nearest neighbour distance.
- Column 15: The mean distance of the conditioning data.
- Column 16: The SID of the nearest conditioning datum.

7.11.3 Gaussim.exe

Input and output formats for *Gaussim.exe* are identical to those of *Geostats.exe*.

7.11.3.i Parameter File

```

FILENAMES [Data,Grid,Output]:
~/ED_ST_Geostats/Data/Data.txt
~/ED_ST_Geostats/Data/Grid.txt
~/ED_ST_Geostats/Outputs/Simulation

METHOD [0=SK,1=OK,2=KED]
2

MODE [0=Debug,1=Default,2=Jack-knife]:
1

SEARCH STRATEGY [Neighbours,Window]:
32      7

STRUCTURES [Spatial,Temporal]:
2      1

TAU [Spatial,Temporal]:
0.02    0.0

GLOBAL SILL:
32.46

COVARIANCE MODEL [0=Product,1=Product-Sum]:
1

SIMULATIONS
1000

Spatial          ]----- Theta -----[ Rotation
=====
Spherical          3.01          9.67          9.67          0.0          0.0
Exponential        6.99         196.55         196.55          0.0          0.0

Temporal          ]----- Theta -----[
=====
Exponential        23.13          6.6           0.0

```

The file is identical to the *Geostats.exe* parameter file, with the addition of the SIMULATIONS heading on line 27. The desired number of realisations from the specified RF should be entered on line 28. The program will automatically append the simulation number and a *.txt* file extension to the filename specified on line 4.

7.12 Acknowledgements

I would like to thank Pierre Goovaerts and Hans Wackernagel for their helpful comments and insights when writing the software.

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8. Discussion

The primary aim of this thesis was to examine and reduce the sources of uncertainty for the estimation of regional to catchment scale carbon budgets. We initially partitioned this task into the influence of the parameterisation of the exchange surface (primarily quantified by LAI), and uncertainties in meteorological driving variables. An investigation of these issues was followed by an analysis of the relative magnitude of uncertainties in the C budget attributable to parameterisation and driver errors respectively. We demonstrated that the dominant source of uncertainty in the final analysis of the C budget was land surface parameterisation, although issues of bias in driver upscaling remain to be resolved.

Land surface parameterisation must be improved to make reliable estimates of C budgets on a regional scale. In Chapter 3 we report persistence of the functional form and approximate error magnitude for LAI NDVI relationships at multiple scales (Williams et al., 2008). However, despite this promising result we demonstrate that it is insufficient to rely on EO derived vegetation indices to provide land surface parameterisations, with weak but significant relationships between key ecosystem variables (LAI) and NDVI (chapters 3 and 4).

On a global scale, DGVMs tend to parameterise the land surface based on a vegetation classification of plant functional types (PFTs) (e.g. Woodward et al., 1995). Results from chapter 4 seem to bear out the validity of this approach. However, the clear variation within PFTs evident in chapter 4 indicates that community dependent topographic relationships may play an important role in regional land surface parameterisation.

Discussion

Approaches based on topographic predictors of LAI were examined in Chapter 4, and appear to offer stronger relationships than EO based approaches. We report at ~50% stronger relationship between LAI and topography than LAI and NDVI. In particular, scale dependent relationships between elevation and shelter (as quantified by TOPEX) are good predictors of LAI.

It is important to remember the dictum that correlation does not imply causation, and whilst statistical relationships between LAI and topography may be useful in a descriptive sense, it is important to develop a mechanistic understanding of the ecophysiology at work in order to successfully implement these findings in an operational sense. It is likely that the large-scale elevation trend is related to temperature, whilst the microscale relationship between LAI and exposure may be explained by local variations in snow accumulation, thaw dates and hence soil nutrient distributions (Wielgolaski et al., 1975). Further experimental work at Abisko is necessary to build process-based understanding of these issues for future modelling work.

We compared various methods of regionalising LAI in the Arctic tundra, to see if statistically optimal interpolation techniques such as Kriging could outperform simpler and computationally cheaper regression techniques. We report broadly similar interpolation skill for various Kriging techniques, inverse distance weighting (IDW) and linear regression, despite the utilisation of different combinations of data streams.

Despite results in chapter 3, where geostatistical methods provided no improvement in interpolation skill over simpler upscaling techniques, geostatistics are likely to remain an important part of any regionalised modelling activity. A common misconception about geostatistical methods is that they are limited to smoothly varying Gaussian fields with a constant mean: Geostatistical methods exist to deal with non-normality (Armstrong and Matheron, 1986a; Armstrong and Matheron, 1986b), discrete

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boundary transitions (Goovaerts and Journel, 1995; Heuvelink and Webster, 2001) and secondary covariates (Goovaerts, 2000; Hudson and Wackernagel, 1994).

Geostatistics provide the ability to assimilate autocorrelated observations into arbitrarily complex regression models, making the techniques an ideal choice for future studies combining PFTs derived from EO vegetation classifications, radiance derived vegetation indices and topographic trends. Mixed effect type models (Pinheiro and Bates, 2000) are readily specified with slight modifications to the standard geostatistical methodologies (see chapter 7.8.2 and 7.9.1.i), providing a framework for the integration of such disparate data sources. Thus geostatistics remain a relevant research area given the significant advantage of offering spatial estimates of parameter variance, which is a key step to developing the potential for spatial data assimilation.

In the second section of the thesis, we turned our attention to problems of upscaling meteorological driving variables. There is a long tradition of such interpolation work in the literature (Ashraf et al., 1997; Hudson and Wackernagel, 1994; Running et al., 1987; Thiessen, 1911; Thornton et al., 1997), although in general this has been attempted at coarser spatio-temporal resolutions (Fuentes et al., 2006). Regionalising precipitation fields is considerably more problematic at fine temporal resolution (Thornton et al., 1997), and we report consistently poor regionalisation of rainfall fields (chapter 5). Geostatistics have previously been reported to provide slight improvements over other methods of regionalisation, and we find broadly similar results to those previously published (Diodato and Ceccarelli, 2005; Goovaerts, 2000): Although geostatistical methods provide an improved analysis over IDW approaches, the relative improvement may be small (Thornton et al., 1997); for example in the case of precipitation our geostatistical analysis provided worse results than IDW.

Discussion

We implemented state of the art spatio-temporal geostatistical techniques to investigate potential improvements to high-resolution meteorological fields over simpler Kriging techniques that ignore temporal autocorrelation. In every case we report a poorer interpolation skill for the spatio-temporally explicit regionalisations (chapter 5). As such, one may question the merit of geostatistical techniques given their perceived complexity, computational cost and marginal improvement in interpolation skill. However, we reiterate the importance of attaching estimates of uncertainty to regionalised variables, which Kriging and its variants (geostatistical techniques subsume spline fitting methods (Matheron, 1981; Serra, 1987)) are uniquely able to achieve.

We examined the effect of *post hoc* temporal data aggregation on estimated fields of meteorology, and report a decrease in error as the size of temporal window increases. This led us to hypothesise that the somewhat large interpolation errors may cancel out over time in the model structure. Specifically, we hypothesised that integrating processes in the model structure would ‘smooth out’ errors, whilst rapid processes which react instantaneously to driving variables would display greater error (chapter 6). We report that temperature was the largest component of the meteorological uncertainty, supporting the hypothesis that instantaneous effects dominate the uncertainty of the NEE trajectory. Despite precipitation having the largest uncertainty and poorest r^2 when compared with observations, the resultant impact on NEE uncertainty was minimal. We demonstrated the temporal buffering of uncertainty attributable to precipitation, and suggest this is because the effect of precipitation on vegetation is manifested through drought; an integrative effect related to soil texture and water holding capacity.

The second aim of the thesis was to compare the magnitude of the effects of parameterisation and driver uncertainty on the total C budget. In chapter 6 we present an analysis with significant novelty, being the first study to implement an Ensemble

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Kalman Filter (ENKF) to provide constrained parameter uncertainty distributions for a C dynamics model. Furthermore it is the first study to implement spatio-temporal geostatistical simulation *via* the product-sum covariance model (De Cesare et al., 2001) to provide an ensemble of probable meteorological scenarios. We report that despite sizable uncertainties in driver fields, only small NEE uncertainties were attributable to meteorology. This was true even under extreme data scarcity, simulated by ignoring all available data <100 km from the study site. Conversely, the uncertainties associated with parameterisation accounted for up to 50% of the total NEE predicted by the model.

The ability of data assimilation to correct parameters based on incoming data streams was demonstrated in Chapter 6, indicating that the tools necessary to constrain and reduce the uncertainties associated with the exchange surface are already in place. The ability of DA techniques to reduce model uncertainty has been proven elsewhere (Evensen, 1994; Quaife et al., 2008; Williams et al., 2005), and such model correction is likely to continue to be important with the arrival of new EO data streams e.g. NASAs Orbiting Carbon Observatory (OCO) (Crisp and Johnson, 2005), and ESAs forthcoming Earth explorer mission (e.g. BIOMASS) (Bensi et al., 2007). It is the opinion of the author that geostatistical and DA technologies are amongst the most promising and relevant areas of C cycle science at present, and are worthy of considerable attention in the future.

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9. Conclusions

The dominant source of uncertainty for regional scale C models is the parameterisation of the land surface. Critically, LAI and foliar N (which may be closely coupled to LAI) must be adequately specified at the working scale to produce reasonable estimates of NEE. EO data sources alone are inadequate to characterise the rapid transitions in LAI typical of Tundra ecosystems and apparent in high-resolution studies. The use of topographic indices derived from high quality DEMs such as those derived by LIDAR may go some way to improving land surface parameterisation. Despite issues of bias in meteorological upscaling, driver uncertainty contributes only marginally to the net uncertainty in C, even in cases of extreme data scarcity. We therefore conclude that future studies should concentrate resources on improving regionalisation of land surface parameters, although an analysis of driver uncertainty is advisable. State of the art spatio-temporal interpolation techniques did not improve driver surface accuracy over those that ignore temporal autocorrelation. We conclude that simpler solutions to upscaling are preferable in terms of computational cost and quality of output. Geostatistical techniques are essential for the calculation of surface error statistics, which are a key step towards regional scale DA implementation; we therefore suggest that regression based techniques or IDW are unsuitable for such studies. DA techniques have proven useful in the correction of model parameterisation, and are likely to provide improvements in model uncertainty in the future, especially if undertaken in 'online' mode such that assimilated observations can adjust parameter trajectories over time.